

Gaurav Raj

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PROFILE

Offensive security researcher and published author specialising in **red team operations, Active Directory / Kerberos, reverse engineering, and kernel development**. Author of the book *Forged Trust* and a full-taxonomy ADCS whitepaper, running penetration testing engagements end to end from client contact through to delivery. Creator of BerserkArch Linux, focused on secure, automated, and scalable offensive security environments. Supported by deep low-level systems experience (C/Assembly/Rust) and Linux internals.

PUBLICATIONS & ACADEMIC CITATIONS

Book: *Forged Trust - Offensive Operations Against ADCS*. Deep fundamentals of ESC1-18, THEFT1-5, and (D)PERSIST.
Paper: *Certificate of Compromise*, a full-taxonomy ADCS SoK / whitepaper. DOI: 10.5281/zenodo.20732733.

TECHNICAL SKILLS

Red Team & Pentest: Reverse Engineering, Network/Internal/Web Pentesting, AD Attacks, C2 (Cobalt Strike/Sliver).
Tooling: NetExec, Impacket, bloodyAD, Sliver C2, BloodHound, Certipy, Ghidra, IDA Pro, BurpSuite Pro.
AI & LLM Security: LLM Fine-tuning, Prompt Engineering, Model Evaluation, Data Analysis, LLM Red Teaming.
Core Languages: C, C++, Assembly, Rust, Golang, Python, Bash, Zig, JavaScript, TypeScript, Java.
Systems Engineering: Linux Internals, Kernel Dev, Memory Management, Bootloaders, Device Drivers, Embedded/IoT.
Cloud & DevOps: AWS, Azure, GCP, Terraform, Vagrant, Kubernetes, Docker, Github Actions, CI/CD, Linux.

SECURITY & ENGINEERING EXPERIENCE

BerserkArch Linux Distro Project <i>Lead Maintainer, Developer & Systems Architect</i>	Remote <i>2025 - Present</i>
OS: Designed and maintain an Arch Linux-based distribution for red team operations and secure development. CI/CD: Built fully automated ISO CI/CD pipelines using GitHub Actions and Docker (build, sign, checksum, deploy). Containerisation: Published official Docker images for portable offensive security labs and rapid development. Security Hardening: Implemented custom kernel patches and AppArmor profiles for system hardening.	

Cybercraft Labs <i>Independent Security Consultant & Engineer</i>	Patna, Bihar <i>Sept 2023 - Nov 2025</i>
Penetration Testing: Delivered network, web, and cloud penetration testing engagements for clients end to end, running each from initial contact and scoping through to reporting and remediation guidance. Cloud Hardening: Hardened client AWS/GCP hybrid infrastructure, including EC2 hardening and SES email authentication (DKIM/DMARC). Software Development: Designed and delivered bespoke client software to specification, including full-stack web applications, desktop tools, and custom reconnaissance automation that streamlined engagement workflows.	

KEY R&D PROJECTS

THBOS (Custom x86 Kernel) <i>Source Code</i>	<i>Systems Programming (C, Assembly & Zig)</i>
64-bit OS kernel from scratch with custom bootloader, Protected Mode, CPUID-driven topology.	
Chip8 Emulator <i>github.com/thehackersbrain/chip8</i>	<i>Systems Emulation (Rust)</i>
Functional CHIP-8 emulator with Full fetch-decode-execute cycle with memory-safe bytecode handling via borrow checker.	
HiddenWave <i>github.com/thehackersbrain/hiddenwave</i>	<i>Steganography Tool (C++ & Python)</i>
LSB audio steganography engine with WAV/MP3 support. Featured in the Netflix series Gyeongseong Creature (S2E1).	

ACHIEVEMENTS

- VulnHub:** Designed 2 vulnerable VMs (*Blogger, CorpVision*) with custom exploit chains (Webmin, Cronjob privesc).
- Advanced Security Labs:** Completed **HackTheBox Fortresses** (Jet) & **TryHackMe Networks** (Wreath, Holo), mastering pivoting, AD exploitation, and perimeter breaching.
- Global Rankings:** **Top 1% on TryHackMe** (CTF) | **HackerRank Gold Badge** (Problem Solving & Algorithms).

EDUCATION

B.Sc (Maths Hons) <i>Specialisation:</i>	<i>Patliputra University, Patna</i> <i>Advanced Mathematics (i.e. Calculus, Linear Algebra)</i>	2027 (ongoing) -
<i>Extensive self-directed study across red team op, adversary simulation, EDR and AV evasion, AD and windows internals.</i>		